
O-RAN Work Group 3 (Near-Real-time RAN Intelligent Controller and E2 Interface)

Y1 interface: Type Definitions

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Foreword

This Technical Specification (TS) has been produced by WG3 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the stage 3 definition of data types that are applicable in the Y1 interface.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE 1: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

NOTE 2: In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in 3GPP Release 17.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.WG3.Y1GAP, "Y1 interface: General Aspects and Principles".
- [2] O-RAN.WG3.Y1AP, "Y1 interface: Application Protocol".
- [3] 3GPP TS 38.314: "NG Application Protocol (NGAP)".
- [4] 3GPP TS 29.571: "5G System; Common Data Types for Service Based Interfaces; Stage 3".
- [5] 3GPP TS 28.552: "Management and orchestration; 5G performance measurements".
- [6] O-RAN.WG3.E2SM, "E2 Service Model (E2SM); KPM".
- [7] 3GPP TS 38.314: "NR; Layer 2 Measurements".
- [8] OpenAPI, "OpenAPI 3.1.0 Specification", <https://github.com/OAI/OpenAPI-Specification/blob/master/versions/3.1.0.md>.
- [9] O-RAN.WG1.OAD, "O-RAN Architecture Description".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE 1: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

NOTE 2: In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in 3GPP Release 17.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [i.1], O-RAN.WG1.OAD [9], O-RAN.WG3.Y1GAP [1] and O-RAN.WG3.Y1AP [2] apply.

3.2 Symbols

For the purposes of the present document, the symbols given in 3GPP TR 21.905 [i.1], O-RAN.WG1.OAD [9], O-RAN.WG3.Y1GAP [1] and O-RAN.WG3.Y1AP [2] apply.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [i.1], O-RAN.WG1.OAD [9], O-RAN.WG3.Y1GAP [1] and O-RAN.WG3.Y1AP [2] apply.

4 General

4.1 Introduction

The data types defined in this document are classified into the following groups:

- Basic data types for generic usage;
- Data types for RAI.

5 Basic data types

5.1 Introduction

This clause defines the basic data types for generic usage.

Table 5.1-1 summarizes the basic data types defined in this specification.

Table 5.1-1: Basic data types defined in this specification

Type Name	Clause	Description	Applicability
PacketDelay	5.3.2	Represents the packet delay.	
PacketLossRate	5.3.2	Represents the packet loss rate.	
Ueid	5.2.2	Identifies a single UE on the Y1 interface.	

Table 5.1-2 summarizes the basic data types re-used from other specifications.

Table 5.1-2: Reused basic data types

Data type	Reference	Comments	Applicability
AmfUeNgapId	3GPP TS 38.413 [3]	AMF UE NGAP ID.	
BitRate	3GPP TS 29.571 [4]	Bit rate.	
Guami	3GPP TS 29.571 [4]	Globally Unique AMF ID.	
Snssai	3GPP TS 29.571 [4]	S-NSSAI.	

5.2 Structured data types

5.2.1 Introduction

This clause defines the structured data types.

5.2.2 Type Ueld

Table 5.2.2-1: Definition of type Ueld

Attribute name	Data type	P	Cardinality	Description	Applicability
amfUeNgapId	AmfUeNgapId	O	0..1	AMF UE NGAP ID. (NOTE 1)	
guami	Guami	C	0..1	Identify the AMF associated with the AMF UE NGAP ID. Shall be present if "amfUeNgapId" attribute is present.	

NOTE 1: In the present document, the "amfUeNgapId" attribute shall be present as no other UE ID has been defined.

5.2.3 Type Distribution

Table 5.2.3-1: Definition of type Distribution

Type Name	Data type	P	Cardinality	Description	Applicability
distribution	array(integer)	M	1..N	This attribute represents the frequency distribution. The array corresponds to the bins of the distribution in ascending order, with each element containing the frequency for each bin.	

5.3 Simple data types and enumerations

5.3.1 Introduction

This clause defines the simple data types and the enumerations.

5.3.2 Simple data types

Table 5.3.2-1: Simple Data Types

Type Name	Type Definition	Description
PacketDelay	integer	Unsigned integer indicating the packet delay as defined in [3], in units of milliseconds.
PacketLossRate	integer	Unsigned integer indicating packet loss rate as defined in [3], in units of tenth of percent. Minimum = 0. Maximum = 1000.

6 RAI specific data types

6.1 Introduction

The data types for RAI are defined per RAI type as specified in clause 6.2 of [1]. Such data types shall be processed according to the RAI type ID and associated RAI type version in the application protocol messages [2].

The RAI types and associated RAI type versions defined in this document are summarized in Table 6.1-1.

Table 6.1-1: RAI types and versions

RAI Type ID	Clause	Version	Supported solution sets
RAN performance analytics	6.2	1.1.0	JSON

Table 6.1-2 summarizes the data types defined per RAI type.

Table 6.1-2: RAI type specific data types

Data type	Description	Reference
RaiTypeId	Indicates the RAI type.	
RaiTypeVersion	Indicates the version of the RAI type.	
FilterParameters	Describes the filter(s) used for the RAI contents.	
RaiContents	Contains the actual RAI.	
TargetEntity	Contains information used to identify a target entity with which the RAI is associated.	
NotificationTriggerEvent	Describes the event that triggers RAI notifications.	

6.2 RAI type: RAN performance analytics

6.2.1 Introduction

This RAI type provides analytics about RAN performance including UE throughput, packet delay, packet loss rate.

The RAI type ID is "RAN performance analytics".

In the present document, the version of the RAI type is "1.1.0".

6.2.2 Data model

6.2.2.1 Introduction

The data types dedicated for RAI type "RAN performance analytics" are specified in the following clauses.

6.2.2.2 Structured data types

6.2.2.2.1 Introduction

This clause defines the structured data types for RAI type "RAN performance analytics".

6.2.2.2.2 Type FilterParameters

Table 6.2.2.2.2-1: Definition of type FilterParameters

Attribute name	Data type	P	Cardinality	Description	Applicability
analyticsSubsetFilters	array(AnalyticsAttributeName)	O	1..N	Indicates the subset of attributes in the RAI contents to be reported.	
binRangeList	array(BinRange)	O	1..N	Indicates the list of bin ranges in ascending order used to generate the distribution for RAI contents.	

6.2.2.2.3 Type RaiContents

Table 6.2.2.2.3-1: Definition of type RaiContents

Attribute name	Data type	P	Cardinality	Description	Applicability
avgRlcThroughput DI	BitRate	O	0..1	The average downlink RLC throughput is defined as the DL Transmitted Data Volume (clause 7.10.1 of [6]) divided by the duration of the validity period, for a specific target entity. (NOTE 1)	
avgRlcThroughput UI	BitRate	O	0..1	The average uplink RLC throughput is defined as the UL Transmitted Data Volume (clause 7.10.2 of [6]) divided by the duration of the validity period, for a specific target entity. (NOTE 1)	
avgPacketDelayDI	PacketDelay	O	0..1	The average downlink packet delay is an overall downlink packet delay in RAN that takes into account the packet delay in the CU-CP (clause 5.1.3.3.1 of [5]), on the F1-U interface (clause 5.1.3.3.2 of [5]), in the DU (clause 5.1.3.3.3 of [5]) and on the air interface (clause 5.1.1.1.1 of [5]), for a specific target entity according to the the conversion rule in clause 7.9.0 of [6].	
avgPacketDelayUI	PacketDelay	O	0..1	The average uplink packet delay is an overall uplink packet delay in RAN that takes into account the packet delay in the CU-CP (clause 5.1.1.1.5 of [5]), in the DU (clause 5.1.1.1.4 of [5]) and on the air interface (clause 5.1.1.1.3 of [5]), for a specific target entity according to the the conversion rule in clause 7.9.0 of [6]. (NOTE 2)	
avgPacketLossRate eDI	PacketLossRate	O	0..1	The average downlink packet loss rate is the overall downlink packet loss rate in RAN that takes into account the packet loss in the CU-CP (clause 5.1.3.2.1 of [5]), on the F1-U interface (clause 5.1.3.1.3 of [5]), in the DU (clause 5.1.3.2.2 of [5]) and on the air interface (clause 4.2.1.5.1 of [7]) during the validity period, for a specific target entity according to the the conversion rule in clause 7.9.0 of [6].	

avgPacketLossRateUI	PacketLossRate	O	0..1	The average uplink packet loss rate is defined as the UL PDCP SDU Loss Rate (clause 5.1.3.1.1 of [5]) during the validity period, for a specific target entity according to the the conversion rule in clause 7.9.0 of [6].	
distPacketDelayDL	Distribution	O	0..1	The distribution of DL packet delay between NG-RAN and UE, which is the delay incurred in NG-RAN (including the delay at gNB-CU-UP, on F1-U and on gNB-DU) and the delay over Uu interface (clause 5.1.1.1.6 of [5]), for a specific target entity according to the conversion rule in clause 7.9.0 of [6]. The unit for the bin ranges of the distribution is milliseconds.	
distPacketDelayUL	Distribution	O	0..1	The distribution of UL packet delay between NG-RAN and UE, which includes the delay occurred in NG-RAN (including the delay at gNB-CU-UP, on F1-U and on gNB-DU) and the delay over Uu interface (excluding the D1 UL PDCP delay occurred in the UE) (clause 5.1.1.1.7.1 of [5]), for a specific target entity according to the conversion rule in clause 7.9.0 of [6]. The unit for the bin ranges of the distribution is milliseconds.	
NOTE 1: Different from the measurements in clause 5.1.1.3.1/5.1.1.3.3 of [5], the downlink/uplink average RLC throughput is calculated regardless of data bursts that require one or multiple slots. NOTE 2: In the present document, the average uplink packet delay does not include the packet delay on the F1-U interface, which is not yet defined.					

6.2.2.2.4 Type TargetEntity

Table 6.2.2.2.4-1: Definition of type TargetEntity

Attribute name	Data type	P	Cardinality	Description	Applicability
ueId	UeId	O	1	Indicates a specific UE. (NOTE 1)	
snssai	Snssai	O	1	Indicate the S-NSSAI.	

In the present document, the allowed combinations of attributes are:

- ueId: the analytics are on a per-UE basis;
- ueId and snssai: the analytics are on a per-UE-per-slice basis.

6.2.2.2.5 Type NotificationTriggerEvent

Table 6.2.2.2.5-1: Definition of type NotificationTriggerEvent

Attribute name	Data type	P	Cardinality	Description	Applicability
n/a				(NOTE 1)	
NOTE 1: In the present document, no notification trigger event has been defined for this RAI type.					

6.2.2.2.6 Type BinRange

Table 6.2.2.2.6-2: Definition of type BinRange

Attribute name	Data type	P	Cardinality	Description	Applicability
startValue	integer	M	1	Indicates the start value of the bin, which corresponds to "Start Value" in clause 8.3.26 of [6].	
endValue	integer	M	1	Indicates the end value of the bin, which corresponds to "End Value" in clause 8.3.26 of [6].	

6.2.2.3 Simple data types and enumerations

6.2.2.3.1 Introduction

This clause defines the simple data types and enumerations for RAI type "RAN performance analytics".

6.2.2.3.2 Simple data types

Table 6.2.2.3.2-1: Simple Data Types

Type Name	Type Definition	Description
n/a		

6.2.2.3.3 Enumeration: AnalyticsAttributeName

Table 6.2.2.3.3-1: Enumeration AnalyticsAttributeName

Enumeration value	Description	Applicability
AVG_RLC_THROUGHPUT_DL	The "avgRlcThroughputDI" attribute for RAI type "RAN performance analytics".	
AVG_RLC_THROUGHPUT_UL	The "avgRlcThroughputUI" attribute for RAI type "RAN performance analytics".	
AVG_PACKET_DELAY_DL	The "avgPacketDelayDI" attribute for RAI type "RAN performance analytics".	
AVG_PACKET_DELAY_UL	The "avgPacketDelayUI" attribute for RAI type "RAN performance analytics".	
AVG_PACKET_LOSS_RATE_DL	The "avgPacketLossRateDI" attribute for RAI type "RAN performance analytics".	
AVG_PACKET_LOSS_RATE_UL	The "avgPacketLossRateUI" attribute for RAI type "RAN performance analytics".	
DIST_PACKET_DELAY_DL	The "distPacketDelayDI" attribute for RAI type "RAN performance analytics".	
DIST_PACKET_DELAY_UL	The "distPacketDelayUI" attribute for RAI type "RAN performance analytics".	

Annex A (normative): OpenAPI specification

A.1 General

This Annex specifies the formal definition of the Y1 data types in this document. It consists of OpenAPI documents in YAML format that are based on the OpenAPI Specification [8].

A.2 RAN performance analytics

```
openapi: 3.1.0

info:
  title: Y1 RAI Type RAN performance analytics
  description: |
    Y1 data types for RAI Type "RAN performance analytics".
    © 2023, O-RAN ALLIANCE
    All rights reserved.
  version: "1.1.0"

externalDocs:
  description: O-RAN.WG3.Y1TD-v01.00
  url: https://www.o-ran.org/specifications

paths: {}

components:
  schemas:
    FilterParameters:
      type: object
      description: Describes the filter(s) used for the RAI contents.
      properties:
        analyticsSubsetFilters:
          type: array
          description: Indicates the subset of attributes in the RAI contents to be reported.
          items:
            $ref: '#/components/schemas/AnalyticsAttributeName'
        binRangeList:
          type: array
          description: Indicates the list of bin ranges in ascending order used to generate the
distribution for RAI contents.
          items:
            $ref: '#/components/schemas/BinRange'

    AnalyticsAttributeName:
      type: string
      enum:
        - AVG_RLC_THROUGHPUT_DL
        - AVG_RLC_THROUGHPUT_UL
        - AVG_PACKET_DELAY_DL
        - AVG_PACKET_DELAY_UL
        - AVG_PACKET_LOSS_RATE_DL
        - AVG_PACKET_LOSS_RATE_UL
        - DIST_PACKET_DELAY_DL
        - DIST_PACKET_DELAY_UL
      description: Indicates an attribute in the RAI contents.

    RaiContents:
      type: object
      description: Contains the actual RAI contents.
      properties:
        avgRlcThroughputDl:
          $ref: '#/components/schemas/BitRate'
```

```

    description: Indicates the average downlink RLC throughput as described in clause 6.2.2.2.3 of
O-RAN.WG3.Y1TD.
    avgRlcThroughputUl:
      $ref: '#/components/schemas/BitRate'
      description: Indicates the average uplink RLC throughput as described in clause 6.2.2.2.3 of O-
RAN.WG3.Y1TD.
    avgPacketDelayDl:
      $ref: '#/components/schemas/PacketDelay'
      description: Indicates the averagedownlink packet delay as described in clause 6.2.2.2.3 of O-
RAN.WG3.Y1TD.
    avgPacketDelayUl:
      $ref: '#/components/schemas/PacketDelay'
      description: Indicates the averageuplink packet delay as described in clause 6.2.2.2.3 of O-
RAN.WG3.Y1TD.
    avgPacketLossRateDl:
      $ref: '#/components/schemas/PacketLossRate'
      description: Indicates the averagedownlink packet loss rate in the air interface as described in
clause 6.2.2.2.3 of O-RAN.WG3.Y1TD.
    avgPacketLossRateUl:
      $ref: '#/components/schemas/PacketLossRate'
      description: Indicates the average uplink packet loss rate as described in clause 6.2.2.2.3 of
O-RAN.WG3.Y1TD.
    distPacketDelayDl:
      $ref: '#/components/schemas/Distribution'
      description: Indicates the distribution of downlink packet delay as described in clause
6.2.2.2.3 of O-RAN.WG3.Y1TD.
    distPacketDelayUl:
      $ref: '#/components/schemas/Distribution'
      description: Indicates the distribution of uplink packet delay as described in clause 6.2.2.2.3
of O-RAN.WG3.Y1TD.

NotificationTriggerEvent:
  type: object
  description: Contains the event that triggers RAI notifications.

BitRate:
  type: string
  pattern: '^\\d+(\\.\\d+)? (bps|Kbps|Mbps|Gbps|Tbps)$'
  description: >
    String representing a bit rate prefixes follow the standard symbols from The International
    System of Units, and represent x1000 multipliers, with the exception that prefix "K" is
    used to represent the standard symbol "k".

PacketDelay:
  type: integer
  minimum: 1
  description: Unsigned integer indicating packet delay expressed in milliseconds.

PacketLossRate:
  type: integer
  minimum: 0
  maximum: 1000
  description: Unsigned integer indicating packet loss rate expressed in tenth of percent.

TargetEntity:
  type: object
  description: Contains information used to identify a target entity with which the RAI is associated.
  properties:
    ueId:
      $ref: '#/components/schemas/UeId'
    snssai:
      $ref: '#/components/schemas/Snssai'

UeId:
  type: object
  description: Indicates a specific UE.
  properties:
    amfUeNgapId:
      $ref: '#/components/schemas/AmfUeNgapId'
    guami:
      $ref: '#/components/schemas/Guami'

```

```

AmfUeNgapiId:
  type: integer
  description: The AMF UE NGAP ID.
  minimum: 0
  maximum: 1099511627775

Guami:
  type: object
  properties:
    plmnId:
      $ref: '#/components/schemas/PlmnIdNid'
    amfId:
      $ref: '#/components/schemas/AmfId'
  required:
    - plmnId
    - amfId
  description: Globally Unique AMF Identifier constructed out of PLMN, Network and AMF identity.

PlmnIdNid:
  type: object
  properties:
    mcc:
      $ref: '#/components/schemas/Mcc'
    mnc:
      $ref: '#/components/schemas/Mnc'
    nid:
      $ref: '#/components/schemas/Nid'
  required:
    - mcc
    - mnc
  description: Contains the serving core network operator PLMN ID and, for an SNPN, the NID that
  together with the PLMN ID identifies the SNPN.

Mcc:
  type: string
  pattern: '^\\d{3}$'
  description: Mobile Country Code part of the PLMN, comprising 3 digits, as defined in clause 9.3.3.5
  of 3GPP TS 38.413.

Mnc:
  type: string
  pattern: '^\\d{2,3}$'
  description: Mobile Network Code part of the PLMN, comprising 2 or 3 digits, as defined in clause
  9.3.3.5 of 3GPP TS 38.413.

Nid:
  type: string
  pattern: '^\\[A-Fa-f0-9\\]{11}$'
  description: This represents the Network Identifier, which together with a PLMN ID is used to
  identify an SNPN (see 3GPP TS 23.003 and 3GPP TS 23.501 clause 5.30.2.1).

AmfId:
  type: string
  pattern: '^\\[A-Fa-f0-9\\]{6}$'
  description: >
    String identifying the AMF ID composed of AMF Region ID (8 bits), AMF Set ID (10 bits)
    and AMF Pointer (6 bits) as specified in clause 2.10.1 of 3GPP TS 23.003. It is encoded
    as a string of 6 hexadecimal characters (i.e., 24 bits).

Snssai:
  type: object
  description: When Snssai needs to be converted to string (e.g. when used in maps as key), the string
  shall be composed of one to three digits "sst" optionally followed by "-" and 6 hexadecimal digits "sd".
  properties:
    sst:
      type: integer
      minimum: 0
      maximum: 255
      description: >
        Unsigned integer, within the range 0 to 255, representing the Slice/Service Type.
        It indicates the expected Network Slice behaviour in terms of features and services.
        Values 0 to 127 correspond to the standardized SST range. Values 128 to 255 correspond

```

to the Operator-specific range. See clause 28.4.2 of 3GPP TS 23.003.
Standardized values are defined in clause 5.15.2.2 of 3GPP TS 23.501.

sd:

type: string

pattern: '^[A-Fa-f0-9]{6}\$'

description: >

3-octet string, representing the Slice Differentiator, in hexadecimal representation. Each character in the string shall take a value of "0" to "9", "a" to "f" or "A" to "F" and shall represent 4 bits. The most significant character representing the 4 most significant bits of the SD shall appear first in the string, and the character representing the 4 least significant bit of the SD shall appear last in the string. This is an optional parameter that complements the Slice/Service type(s) to allow to differentiate amongst multiple Network Slices of the same Slice/Service type. This IE shall be absent if no SD value is associated with the SST.

required:

- sst

Distribution:

type: array

description: Indicates the frequency distribution. The array corresponds to the bins of the distribution in ascending order, with each element containing the frequency for each bin.

items:

type: integer

BinRange:

type: object

description: Indicates the value range of a bin of the distribution.

properties:

startValue:

type: integer

description: Indicates the start value of the bin, which corresponds to "Start Value" defined in clause 8.3.26 of E2SM KPM.

endValue:

type: integer

description: Indicates the end value of the bin, which corresponds to "End Value" defined in clause 8.3.26 of E2SM KPM.

Annex (informative): Change History

Date	Revision	Description
2023.09.04	00.00.01	Create the skeleton.
2023.10.24	00.00.02	Implemented CMCC-2023.09.26-WG3-CR-0002-Y1TD-Basic contents for Y1TD-v02.docx and editorial changes.
2023.11.17	00.00.03	Editorial changes according to the review comments during WG3 poll.
2023.11.18	01.00	Version incremented for publication.
2024.07.08	01.00.01	Implemented KDDI-2024.02.15-WG3-CR-0002-Y1TD-Enhancement for delay attributes-v02.docx KDDI-2024.02.15-WG3-CR-0004-Y1TD-Adding attributes for setting bin ranges for distribution-v02.docx CMCC-2024.05.27-WG3-CR-0003-Y1TD-Editorial corrections for ODR alignment-v03.docx and editorial changes.
2024.07.22	01.00.02	Editorial changes according to the review comments during WG3 poll.
2024.07.26	01.00.03	Incremented OpenAPI code version to 1.1.0.
2024.07.26	02.00	Version incremented for TSC.